GRADE 9 MATHEMATICS – LESSON PLAN 1

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Orientation / Consolidation

Sub-strand: Introduction and diagnostic assessment

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recall prerequisite knowledge, identify learning needs and set personal improvement goals.
- **Skills:** The learner should be able to apply the concepts and procedures in introduction and diagnostic assessment to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate confidence, accuracy and perseverance when working mathematically.

Key Inquiry Question: What strategies help us identify strengths and areas for improvement in Mathematics?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to introduction and diagnostic assessment. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics?

30 minutes – Lesson Development

Step 1: The learner is guided to explore introduction and diagnostic assessment using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 2

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Orientation / Consolidation

Sub-strand: Introduction and diagnostic assessment

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recall prerequisite knowledge, identify learning needs and set personal improvement goals.
- **Skills:** The learner should be able to apply the concepts and procedures in introduction and diagnostic assessment to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate confidence, accuracy and perseverance when working mathematically.

Key Inquiry Question: What strategies help us identify strengths and areas for improvement in Mathematics?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to introduction and diagnostic assessment. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics?

30 minutes – Lesson Development

Step 1: The learner is guided to explore introduction and diagnostic assessment using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 3

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Orientation / Consolidation

Sub-strand: Introduction and diagnostic assessment

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recall prerequisite knowledge, identify learning needs and set personal improvement goals.
- **Skills:** The learner should be able to apply the concepts and procedures in introduction and diagnostic assessment to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate confidence, accuracy and perseverance when working mathematically.

Key Inquiry Question: What strategies help us identify strengths and areas for improvement in Mathematics?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to introduction and diagnostic assessment. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics?

30 minutes – Lesson Development

Step 1: The learner is guided to explore introduction and diagnostic assessment using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 4

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Orientation / Consolidation

Sub-strand: Introduction and diagnostic assessment

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recall prerequisite knowledge, identify learning needs and set personal improvement goals.
- **Skills:** The learner should be able to apply the concepts and procedures in introduction and diagnostic assessment to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate confidence, accuracy and perseverance when working mathematically.

Key Inquiry Question: What strategies help us identify strengths and areas for improvement in Mathematics?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to introduction and diagnostic assessment. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics?

30 minutes – Lesson Development

Step 1: The learner is guided to explore introduction and diagnostic assessment using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What strategies help us identify strengths and areas for improvement in Mathematics? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 5

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 6

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 7

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 8

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 9

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 10

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Integers

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use integers in real-life contexts.
- **Skills:** The learner should be able to apply the concepts and procedures in integers to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show confidence, persistence and accuracy when solving problems involving positive and negative quantities.

Key Inquiry Question: How do integers represent quantities above and below a reference point?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to integers. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do integers represent quantities above and below a reference point?

30 minutes – Lesson Development

Step 1: The learner is guided to explore integers using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do integers represent quantities above and below a reference point? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 11

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 12

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 13

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 14

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 15

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 16

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Cubes and Cube Roots

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to identify perfect cubes and determine cube roots.
- **Skills:** The learner should be able to apply the concepts and procedures in cubes and cube roots to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value logical reasoning, patience and checking the reasonableness of answers.

Key Inquiry Question: How can the structure of a cube help us determine its volume and cube root?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to cubes and cube roots. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can the structure of a cube help us determine its volume and cube root?

30 minutes – Lesson Development

Step 1: The learner is guided to explore cubes and cube roots using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can the structure of a cube help us determine its volume and cube root? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 17

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 18

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 19

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 20

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 21

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 22

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 23

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 24

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Indices and Logarithms

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply laws of indices and introductory logarithmic relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in indices and logarithms to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate curiosity, accuracy and perseverance when simplifying expressions.

Key Inquiry Question: How do indices and logarithms make very large or small numbers easier to express?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to indices and logarithms. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do indices and logarithms make very large or small numbers easier to express?

30 minutes – Lesson Development

Step 1: The learner is guided to explore indices and logarithms using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do indices and logarithms make very large or small numbers easier to express? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 25

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 26

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 27

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 28

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 29

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 30

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 31

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 32

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 33

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Numbers

Sub-strand: Compound Proportions and Rates of Work

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving compound proportion and rates of work.
- **Skills:** The learner should be able to apply the concepts and procedures in compound proportions and rates of work to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate teamwork, fairness and efficient use of time and resources.

Key Inquiry Question: How can proportional reasoning help us plan work and use resources efficiently?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to compound proportions and rates of work. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently?

30 minutes – Lesson Development

Step 1: The learner is guided to explore compound proportions and rates of work using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can proportional reasoning help us plan work and use resources efficiently? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 34

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 35

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 36

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 37

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 38

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 39

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 40

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 41

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Matrices

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent information using matrices and perform the operations required in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in matrices to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show neatness, orderliness and accuracy when organising and presenting information.

Key Inquiry Question: How can matrices organise information and support problem-solving?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to matrices. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can matrices organise information and support problem-solving?

30 minutes – Lesson Development

Step 1: The learner is guided to explore matrices using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can matrices organise information and support problem-solving? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 42

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 43

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 44

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 45

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 9	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 46

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 47

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 48

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 49

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 50

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 10	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 51

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 11	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 52

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 11	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 53

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 11	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 54

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 11	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 55

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 1	WEEK: 11	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 56

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Equation of a Straight Line

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to represent and determine equations of straight lines.
- **Skills:** The learner should be able to apply the concepts and procedures in equation of a straight line to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision, logical reasoning and the usefulness of graphs in everyday situations.

Key Inquiry Question: What information about a relationship can the equation and graph of a straight line reveal?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to equation of a straight line. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal?

30 minutes – Lesson Development

Step 1: The learner is guided to explore equation of a straight line using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What information about a relationship can the equation and graph of a straight line reveal? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 57

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 58

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 59

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 60

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 61

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 62

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Algebra

Sub-strand: Linear Inequalities

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to form, represent and solve linear inequalities.
- **Skills:** The learner should be able to apply the concepts and procedures in linear inequalities to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate open-mindedness and justify decisions using evidence and mathematical reasoning.

Key Inquiry Question: How are inequalities useful when a situation has limits or conditions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to linear inequalities. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are inequalities useful when a situation has limits or conditions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore linear inequalities using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are inequalities useful when a situation has limits or conditions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 63

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 64

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 65

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 66

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 67

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 68

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 69

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 70

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Area

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to calculate areas of relevant plane figures and composite shapes.
- **Skills:** The learner should be able to apply the concepts and procedures in area to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value accuracy and careful measurement in practical situations.

Key Inquiry Question: How can a complex shape be divided into simpler shapes to find its area?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to area. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can a complex shape be divided into simpler shapes to find its area?

30 minutes – Lesson Development

Step 1: The learner is guided to explore area using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can a complex shape be divided into simpler shapes to find its area? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 71

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 72

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 73

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 74

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 75

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 76

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 77

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 78

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Volume of Solids

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to determine volumes of solids covered in the grade-level design.
- **Skills:** The learner should be able to apply the concepts and procedures in volume of solids to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and responsible use of materials when solving spatial problems.

Key Inquiry Question: How do dimensions affect the space occupied by a solid?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to volume of solids. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do dimensions affect the space occupied by a solid?

30 minutes – Lesson Development

Step 1: The learner is guided to explore volume of solids using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do dimensions affect the space occupied by a solid? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 79

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 80

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 81

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 82

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 83

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 84

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 85

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 86

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Mass, Volume, Weight and Density

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to distinguish mass and weight and relate density to mass and volume.
- **Skills:** The learner should be able to apply the concepts and procedures in mass, volume, weight and density to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate care, honesty and safety when taking and recording measurements.

Key Inquiry Question: How can we use mass and volume to compare materials?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to mass, volume, weight and density. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can we use mass and volume to compare materials?

30 minutes – Lesson Development

Step 1: The learner is guided to explore mass, volume, weight and density using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can we use mass and volume to compare materials? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 87

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 88

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 89

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 90

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 91

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 92

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 93

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 94

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 95

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 96

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 9	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Time, Distance and Speed

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to use relationships among time, distance and speed to solve problems.
- **Skills:** The learner should be able to apply the concepts and procedures in time, distance and speed to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value punctuality, accuracy and safe decision-making in movement and travel.

Key Inquiry Question: How are time, distance and speed related in everyday movement?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to time, distance and speed. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How are time, distance and speed related in everyday movement?

30 minutes – Lesson Development

Step 1: The learner is guided to explore time, distance and speed using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How are time, distance and speed related in everyday movement? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 97

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 9	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 98

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 9	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 99

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 9	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 100

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 9	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 101

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 10	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 102

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 10	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 103

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 10	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Money

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to solve problems involving money, prices, budgeting and financial decisions within the curriculum scope.
- **Skills:** The learner should be able to apply the concepts and procedures in money to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate honesty, prudence and responsibility when making financial decisions.

Key Inquiry Question: How can mathematics support responsible financial decisions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to money. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can mathematics support responsible financial decisions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore money using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can mathematics support responsible financial decisions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 104

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 10	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Approximations and Errors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to round and estimate appropriately.
- **Skills:** The learner should be able to apply the concepts and procedures in approximations and errors to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and acknowledge the limitations of measurements.

Key Inquiry Question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to approximations and errors. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

30 minutes – Lesson Development

Step 1: The learner is guided to explore approximations and errors using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 105

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 10	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Approximations and Errors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to round and estimate appropriately.
- **Skills:** The learner should be able to apply the concepts and procedures in approximations and errors to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and acknowledge the limitations of measurements.

Key Inquiry Question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to approximations and errors. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

30 minutes – Lesson Development

Step 1: The learner is guided to explore approximations and errors using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 106

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 11	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Approximations and Errors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to round and estimate appropriately.
- **Skills:** The learner should be able to apply the concepts and procedures in approximations and errors to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and acknowledge the limitations of measurements.

Key Inquiry Question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to approximations and errors. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

30 minutes – Lesson Development

Step 1: The learner is guided to explore approximations and errors using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 107

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 11	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Measurements

Sub-strand: Approximations and Errors

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to round and estimate appropriately.
- **Skills:** The learner should be able to apply the concepts and procedures in approximations and errors to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate accuracy and acknowledge the limitations of measurements.

Key Inquiry Question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to approximations and errors. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy?

30 minutes – Lesson Development

Step 1: The learner is guided to explore approximations and errors using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: Why should measurements and calculated answers be reported to an appropriate degree of accuracy? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 108

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 11	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 109

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 11	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 110

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 2	WEEK: 11	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 111

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 1	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 112

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 1	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 113

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 1	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Coordinates and Graphs

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to locate and interpret points on coordinate axes and use graphs to represent relationships.
- **Skills:** The learner should be able to apply the concepts and procedures in coordinates and graphs to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show precision and confidence when locating and interpreting positions.

Key Inquiry Question: How do coordinates help us describe position and relationships?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to coordinates and graphs. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How do coordinates help us describe position and relationships?

30 minutes – Lesson Development

Step 1: The learner is guided to explore coordinates and graphs using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How do coordinates help us describe position and relationships? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 114

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 1	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 115

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 1	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 116

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 2	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 117

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 2	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 118

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 2	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 119

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 2	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 120

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 2	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 121

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 3	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 122

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 3	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 123

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 3	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 124

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 3	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 125

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 3	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 126

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 4	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 127

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 4	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Scale Drawing

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to interpret and construct scale drawings.
- **Skills:** The learner should be able to apply the concepts and procedures in scale drawing to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value neatness, accuracy and attention to detail in representing real objects.

Key Inquiry Question: How does a scale drawing represent a real object or place accurately?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to scale drawing. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How does a scale drawing represent a real object or place accurately?

30 minutes – Lesson Development

Step 1: The learner is guided to explore scale drawing using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How does a scale drawing represent a real object or place accurately? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 128

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 4	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 129

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 4	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 130

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 4	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 131

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 5	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 132

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 5	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 133

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 5	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 134

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 5	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 135

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 5	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Similarity and Enlargement

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to recognise similar figures and perform enlargements using scale factors.
- **Skills:** The learner should be able to apply the concepts and procedures in similarity and enlargement to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should appreciate precision and recognise the usefulness of scale in design.

Key Inquiry Question: What remains the same and what changes when a shape is enlarged?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to similarity and enlargement. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What remains the same and what changes when a shape is enlarged?

30 minutes – Lesson Development

Step 1: The learner is guided to explore similarity and enlargement using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What remains the same and what changes when a shape is enlarged? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 136

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 6	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 137

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 6	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 138

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 6	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 139

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 6	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 140

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 6	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 141

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 7	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 142

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 7	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Geometry

Sub-strand: Trigonometry

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to apply the trigonometric ratios within the grade-level scope to solve right-angled triangle problems.
- **Skills:** The learner should be able to apply the concepts and procedures in trigonometry to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should demonstrate persistence and accuracy when solving problems involving right-angled triangles.

Key Inquiry Question: How can trigonometric ratios help determine an unknown length or angle?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to trigonometry. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can trigonometric ratios help determine an unknown length or angle?

30 minutes – Lesson Development

Step 1: The learner is guided to explore trigonometry using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can trigonometric ratios help determine an unknown length or angle? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 143

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 7	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 144

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 7	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 145

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 7	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 146

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 8	LESSON: 1	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 147

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 8	LESSON: 2	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 148

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 8	LESSON: 3	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Data Interpretation (Grouped Data)

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to organise, read and interpret grouped data.
- **Skills:** The learner should be able to apply the concepts and procedures in data interpretation (grouped data) to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should value honesty, objectivity and responsibility when interpreting and communicating data.

Key Inquiry Question: What can grouped data tell us, and what limitations should we consider?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to data interpretation (grouped data). Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: What can grouped data tell us, and what limitations should we consider?

30 minutes – Lesson Development

Step 1: The learner is guided to explore data interpretation (grouped data) using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: What can grouped data tell us, and what limitations should we consider? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 149

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 8	LESSON: 4	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Probability

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe likelihood and determine probabilities of simple events within the grade-level scope.
- **Skills:** The learner should be able to apply the concepts and procedures in probability to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show curiosity and make reasoned judgements when dealing with uncertainty.

Key Inquiry Question: How can probability help us describe uncertainty and make predictions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to probability. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can probability help us describe uncertainty and make predictions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore probability using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can probability help us describe uncertainty and make predictions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

GRADE 9 MATHEMATICS – LESSON PLAN 150

SCHOOL: _____	LEVEL: Grade 9	LEARNING AREA: Mathematics	DATE: _____
TERM: 3	WEEK: 8	LESSON: 5	TIME: 40 minutes | ROLL: _____

Strand: Data Handling and Probability

Sub-strand: Probability

Lesson Learning Outcomes

At the end of the lesson, the learner should be able to:

- **Knowledge:** The learner should be able to describe likelihood and determine probabilities of simple events within the grade-level scope.
- **Skills:** The learner should be able to apply the concepts and procedures in probability to solve relevant mathematical and real-life problems, showing clear working and checking answers.
- **Attitude:** The learner should show curiosity and make reasoned judgements when dealing with uncertainty.

Key Inquiry Question: How can probability help us describe uncertainty and make predictions?

Learning Resources: Learner's book; teacher-prepared notes; charts; number line/graph paper; geometric instruments; locally available materials; calculator where appropriate.

Organization of Learning: Learners work individually, in pairs and in small groups as appropriate. They share mathematical thinking, use concrete or visual representations where suitable, and complete independent practice. Provide differentiated prompts and support according to learner readiness.

Lesson Development

5 minutes – Introduction

Connect the lesson to learners' prior knowledge and an everyday context related to probability. Present a short problem, object, diagram or number example. Learners share initial ideas, make predictions and identify what they need to find out. Introduce the key inquiry question: How can probability help us describe uncertainty and make predictions?

30 minutes – Lesson Development

Step 1: The learner is guided to explore probability using relevant examples and available learning resources. Learners record their working and discuss the strategy used.

Step 2: work out examples individually and in pairs. Learners record their working and discuss the strategy used.

Step 3: draw, construct or interpret mathematical representations where appropriate. Learners record their working and discuss the strategy used.

Step 4: discuss strategies and compare solutions in groups. Learners record their working and discuss the strategy used.

Extended Activities

Learners who complete the core task investigate a more challenging or real-life application of the same concept, create a question for classmates, or use an alternative representation to explain their solution. Learners needing support use worked examples, number cards, diagrams or guided practice.

5 minutes – Conclusion

Learners summarise the main idea in their own words and respond to the key inquiry question: How can probability help us describe uncertainty and make predictions? They complete a brief exit question or explain one method they used. The teacher checks understanding and identifies any concept requiring reinforcement.

Assessment

Suggested assessment evidence: Oral questions; observation; class exercises; written tasks; peer/self-assessment; practical task; exit ticket.

Reflection in the lesson / self-remarks

To be completed after teaching: record learners who achieved the outcomes, evidence collected, misconceptions or errors observed, learners requiring additional support, and the next instructional step.

Implementation Notes

This collection contains 150 Grade 9 Mathematics lesson plans organised according to the three-term full-year scheme supplied for this project. Each plan follows the supplied lesson-plan structure and uses the scheme's strand, sub-strand, outcomes, learning experiences, key inquiry question, resources and assessment methods.

The 40-minute lesson structure is presented as 5 minutes for introduction, 30 minutes for lesson development and 5 minutes for conclusion. Adjust timing to the actual school timetable and learner needs.

Lesson activities are planned to be learner-centred. The teacher facilitates, observes, questions and provides differentiated support. Complete the reflection section after each lesson.

This is a teacher-prepared resource. Check the lesson sequence against the current KICD Grade 9 Mathematics Curriculum Design and the school's approved scheme before classroom implementation.